

LUYAO NIU (SHE/HER)

PhD student, University of Southern California
luyaoniu@usc.edu | [Academic homepage](#)

EDUCATION

University of Southern California

PhD · Transportation Systems Engineering | 2026.08–present

Peking University

M.Sc. · Urban Planning / Smart City & Big Data | 2023–2026

Beijing Jiaotong University

B.Sc. · Traffic Engineering; minor in Finance | 2019–2023

RESEARCH INTERESTS

Learning, decision-making, and coordination in intelligent systems, with a growing focus on multi-agent and autonomous systems.

Learning & Reasoning | Multi-Agent Decision-Making | Coordination & Collective Intelligence

Current questions

Coordination in autonomous systems, decisions under imperfect information, and the role of personal history in mobility intelligence.

Coordination & resilient mobility

How can interacting fleets and institutions coordinate in disrupted transportation networks?

Multi-agent systems · Game theory · Network optimization

Information & autonomous decisions

How can autonomous systems reason and act when sensing and communication are incomplete?

Information design · Uncertainty modeling · Decision-making

Personal history & behavioral intelligence

When does personal history help us understand behavior and make useful predictions?

Representation learning · Evaluation design · Causal analysis

PROFESSIONAL EXPERIENCE

Agentic AI Engineer · Intern | May - Aug. 2026

rednote (Xiaohongshu) · Data Analytics · Shanghai, China · On-site

- Domain knowledge infrastructure: Built a domain-specific data analytics knowledge base to support knowledge-grounded agent workflows.
- Agent architecture and development: Developed agent frameworks and data-analysis workflows that connect knowledge resources with domain capabilities.
- Validation and Skill operations: Validated end-to-end agent behavior and analytical outputs; designed, evaluated, monitored, and operated reusable domain Skills.

Co-founder | Dec. 2024 - Present

Qianmo Zhilian (Shenzhen) Technology Co., Ltd.

- AskNearby: Co-founded and led development of an LLM-powered community information platform combining RAG and spatio-temporal data.
- Recognition: The project received a Gold Award at the Nanshan District Innovation & Entrepreneurship Competition and Third Prize at the Baidu ERNIE Cup.

Research Assistant | Jan. 2025 - Jun. 2026

Duke Kunshan University · Internship · Suzhou, China · Remote

- Research leadership: Led independent research on spatio-temporal data mining, traffic forecasting, and behavioral modeling.
- Research output: First-authored work published at IEEE GLOBECOM; Event-CausNet submitted to KDD 2027.

Graduate Research Assistant | Sep. 2023 - Jun. 2026

Peking University · Full-time · Beijing, China · On-site

Management Intern | Sep. 2022 - Nov. 2022

China Academy of Railway Sciences · Haidian, Beijing, China · On-site

LLM Algorithm R&D | Apr. 2025 - Jan. 2026

China Academy of Land Surveying and Planning · Collaborative project

- Knowledge and reporting agents: Led the RAG retrieval module for a policy knowledge base and developed an LLM-agent reporting workflow.
- Workflow impact: The reporting workflow reduced initial-draft preparation time by more than 90%.

ACADEMIC SERVICE

Journal reviewing

- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Transactions on Big Data (TBD)

Conference reviewing

- ACM MSWiM
- IEEE GLOBECOM
- Transportation Research Board (TRB)

ORAL PRESENTATIONS

IEEE GLOBECOM 2025 | December 2025 | Taipei

ACM SIGSPATIAL GeoAI 2025 | November 2025 | Minneapolis, USA

EUGEO 2025 | September 2025 | Vienna, Austria

AESOP 2025 | July 2025 | Istanbul, Türkiye

HONORS & AWARDS

Scholarships & honors

- Baosteel Outstanding Student Award (500 recipients nationwide / year), 2022.12
- National Scholarship (highest undergraduate honor in China, top 1%), 2020.12
- First-class Academic Scholarship (top 3%), 2020.12
- Beijing Outstanding Graduate

Competition awards

- First Prize, National University Traffic Science & Technology Competition, 2022.07
- Second Prize, China MathorCup Mathematical Modeling Competition, 2022.04
- Third Prize, China College Mathematics Competition, 2020.04
- Third Prize, Baidu ERNIE Cup AI Entrepreneurship Competition, 2024.09
- Gold Award, Nanshan District Innovation & Entrepreneurship Competition

LEADERSHIP & SERVICE

President, Student Union | Beijing Jiaotong University | Jun. 2021 - Jun. 2023

Led university-wide student affairs and brand events reaching 80,000+ participants.

Volunteer, Beijing 2022 Winter Olympics | National Indoor Stadium | Jan. 2022 - Apr. 2022

Personnel management and venue operations; named Outstanding Volunteer.

New Youth Global Competence Development Program | 2022–2023

China Education Association for International Exchange (CEAIE) and partner organizations

Participated in a program focused on international perspectives, global competence, and intercultural communication.

PUBLICATIONS & MANUSCRIPTS

Journal

Niu, C.; He, S.; Fang, C.; **Niu, L.**; Xiao, Y.; Zhang, W. (2026). Gender and Age Disparities in Public Health Crisis Disruptions to Cross-Border Mobility: A Causal Analysis of Daily Flows Between Shenzhen and Hong Kong. *Systems*, 14(7), 772. <https://doi.org/10.3390/systems14070772>

Shi, Z.; Luo, S.; **Niu, L.**; Shao, C. (2025). Joint Design of Public Transit and Autonomous Vehicle Mobility Systems for Tourist Cities Considering Abnormal Trips. *Transportation Research Part C*. <https://doi.org/10.1016/j.trc.2025.105179>

Liu, C.; **Niu, L.**; Zhang, W. (2025). Rebound or Substitution? The Causal Effect of Electric Vehicle Adoption on Individual Travel Behaviour: A Quasi-Experiment. *Applied Geography* (Under review).

Conference

Wang, Z.*; **Niu, L.***; Li, R.; Ban, J. (X.). (2027). GeoSplit: Selective Edge–Server Map Acquisition for Cross-User Travel-Mode Recognition. ICASSP 2027 (Submitted).

** Equal contribution.*

Wang, Z.*; Liao, B.*; **Niu, L.**; Li, R.; Ban, J. (X.). (2027). MSDLoss: EMA-Adaptive Multi-Scale Difference Loss for Time Series Forecasting. ICASSP 2027 (Submitted).

** Equal contribution.*

Niu, L.; Deng, Z.; Li, B.; Huang, N.; Liu, R.; Zhang, W. (2025). AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations. ACM SIGSPATIAL GeoAI '25. <https://doi.org/10.48550/arXiv.2512.02502>

Niu, L.; Wang, Z.; Liu, J.; Boukerche, A.; Sun, P. (2025). MF-AttnBiLSTM: Traffic Flow Prediction via Hybrid Signal Decomposition and Dual-Stream Temporal Attention Learning. GLOBECOM 2025 — 2025 IEEE Global Communications Conference, pp. 2952–2957. IEEE. <https://ieeexplore.ieee.org/abstract/document/11432074/>

Niu, L.; Wang, Y.; He, S.; Zhang, W. (2025). Evaluating the Causal Impacts on Spatial Satisfaction Sentiment of Shenzhen–Hong Kong Regional Integration Policies: An Interrupted Time Series Study Leveraging LLMs. EUGEO 2025.

Niu, L.; Gu, J.; Liu, C.; Zhang, W. (2025). Why Public Transport Policies Work or Not? A Meta-analysis Based on the Behaviour Change Wheel. AESOP 2025.

Manuscripts & Preprints

Niu, L.; Li, R. (2026). Blind Spots: How Uneven Information Shapes Competing Autonomous Fleets and Where to Place Public Sensing. Working manuscript.

Niu, L.; Li, R. (2026). Protecting Critical Roads during Wildfires: Selective Coordination of Commercial AV Fleets. Working manuscript.

Niu, L.; Li, R. (2026). Fleet Coordination for Braess Recovery: Organization and Penetration Thresholds on a Fixed Network. Working manuscript.

Tsao, M.; Guan, Y.; **Niu, L.**; Li, R. (2026). Operational Competition in Robotaxi Systems with Congestion and Heterogeneous Constraints. Working manuscript.

Niu, L.; Wang, Y.; Niu, C.; He, S.; Cai, Y.; Zhang, W. (2026). Designation is not experience: Visitor–host divergence in cross-border spatial satisfaction after integration policies in the Shenzhen–Hong Kong corridor, measured with large language models and calibrated interrupted time series. Working manuscript.

Niu, L.; Wang, Z.; Guan, S.; Sun, P. (2025). Event-CausNet: Unlocking Causal Knowledge from Text with LLMs for Reliable Spatio-Temporal Forecasting. KDD 2027 (Submitted). <https://doi.org/10.48550/arXiv.2511.12769>

Niu, L.; Huang, N. (2025). ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification. arXiv (Preprint). <https://doi.org/10.48550/arXiv.2511.13702>

Niu, L.; Wang, Y.; He, S.; Zhang, W. (2025). Leveraging Large Language Models to Measure Cross-Border Spatial Identity from Multilingual Social Media Data. *Transactions in GIS* (Under review).

Niu, L.; Xing, X.; Liu, Z.; Yao, E. (2025). A Traffic Assignment Model Balancing User Preference and System Optimum. In preparation.

Zhu, H.; Zhang, W.; Huang, N.; Li, B.; **Niu, L.**; ...; Liu, X. (2024). PlanGPT: Enhancing Urban Planning with Tailored Language Model and Efficient Retrieval. arXiv (Preprint). <https://doi.org/10.48550/arXiv.2402.19273>